Michael Remington Coulter

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PROFESSIONAL EXPERIENCE

Arm | Austin, TX

Senior Software Engineer

Apr 2023 - Present

Develop Interconnect Architect, Arm's Python/TypeScript SoC interconnect configuration tool for internal engineers and external customers, alongside AI tools for engineering workflows.

- Developed a Rust/Tauri 2 workbench for repository, build and process control, with a React/TypeScript UI, shared desktop/CLI core and bounded asynchronous Git inspection.
- Built Python Model Context Protocol (MCP) tools for agent-driven interconnect configuration, with typed plans, checkpointed execution, semantic validation and rollback.
- Validated creation, import, modification, diagnosis and rollback in two bounded agent evaluations; rollback restored the exact configuration hash. 92 live acceptance checks passed.
- CMN GUI Lead: created and own all coherent-mesh GUI tooling from CMN-800 onward, translating hierarchical hardware models into Angular/TypeScript editors and diagnostics.
- Created the NoC topology editor and configuration/diagnostic console; implemented route-aware layout and shared interaction behavior across SVG and Pixi.js renderers.
- Introduced Pixi.js and optimized Angular interactions for NoC canvas performance; created Playwright infrastructure and guarded against stale validation results.
- Built VS Code launch/test tools with exact Playwright test-ID verification, plus a multi-root navigation extension with stale-result protection.
- Automated Jenkins/LSF failure replay at the original revision, patch transfer and repair validation; gated flaky-test fixes on three consecutive clean replays.

Recognition: Most Innovative Award, Engineering AI Showcase (Dec 2025): multi-agent orchestration and context-gathering system.

Software Engineer

Apr 2020 - Apr 2023

- Built Flask services for Jupyter notebook lifecycle and AWS EFS persistence in Pelion's embedded-ML platform; designed PostgreSQL schemas and end-to-end tests.

Graduate Software Engineer

Jan 2019 - Apr 2020

- Built Verilog FPGA modules and Mbed OS firmware to test board I/O; contributed to test-shield PCBs. Arm Quality Award (team, 2020).

TECHNICAL SKILLS

Languages: TypeScript/JavaScript, Python, Rust, SQL
Applications: React, Angular, RxJS, Electron, Tauri 2, Flask, Node.js, PostgreSQL
AI / developer tools: MCP, agent evaluation, Git, VS Code extensions, Playwright, Karma, Jenkins

EDUCATION

The University of Texas at Austin

Bachelor of Science, Electrical and Computer Engineering | Dec 2018
Primary focus: Computer Architecture and Embedded Systems | Secondary: Integrated Circuits